Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following task to reduce the amount of network traffic viewed in the simulation.

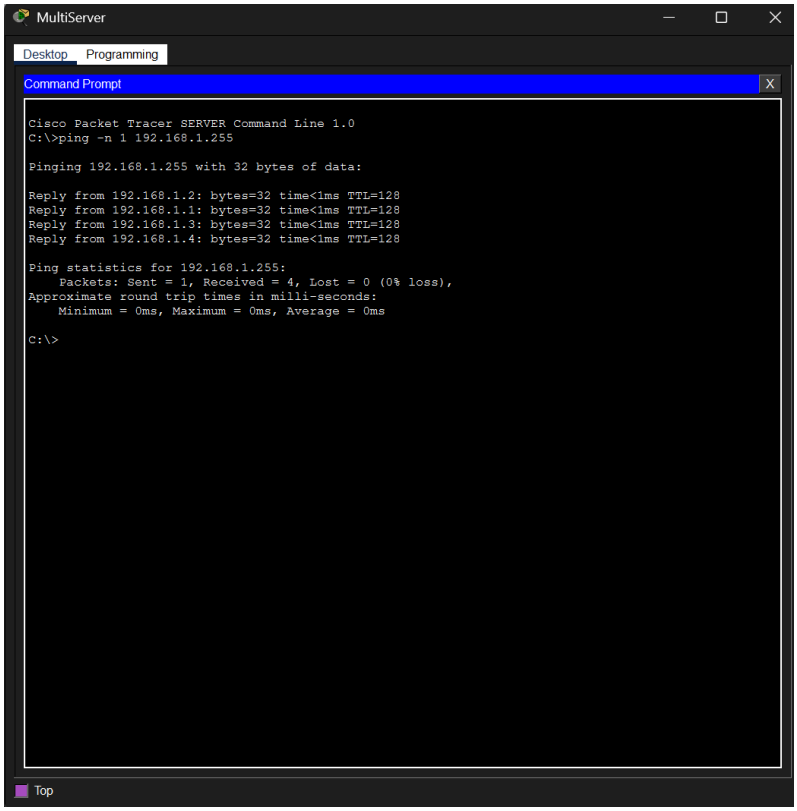
a. Click MultiServer and click the Desktop tab > Command Prompt.

b. Enter the ping -n 1 192.168.1.255 command. You are pinging the broadcast address for the client LAN.

The command option will send only one ping request rather than the usual four. This will take a few

seconds as every device on the network responds to the ping request from MultiServer.

c. Close the MultiServer window.



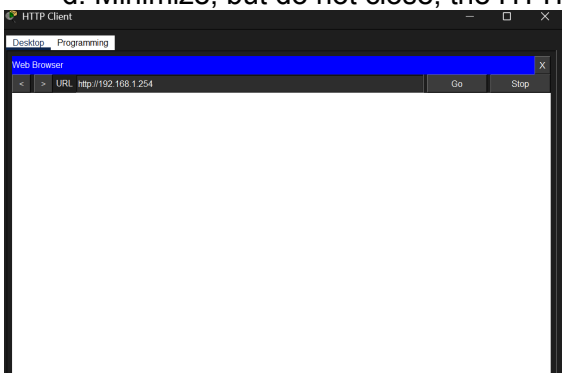
Step 2: Generate web (HTTP) traffic.

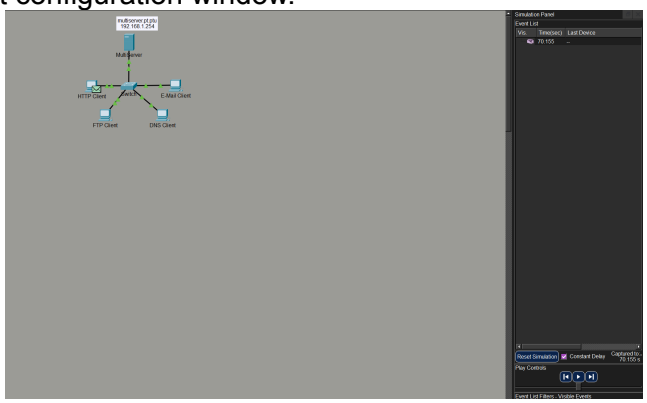
a. Switch to Simulation mode.

b. Click HTTP Client and open the Web Browser from the desktop.

c. In the URL field, enter 192.168.1.254 and click Go. Envelopes (PDUs) will appear in the topology window.

d. Minimize, but do not close, the HTTP Client configuration window.



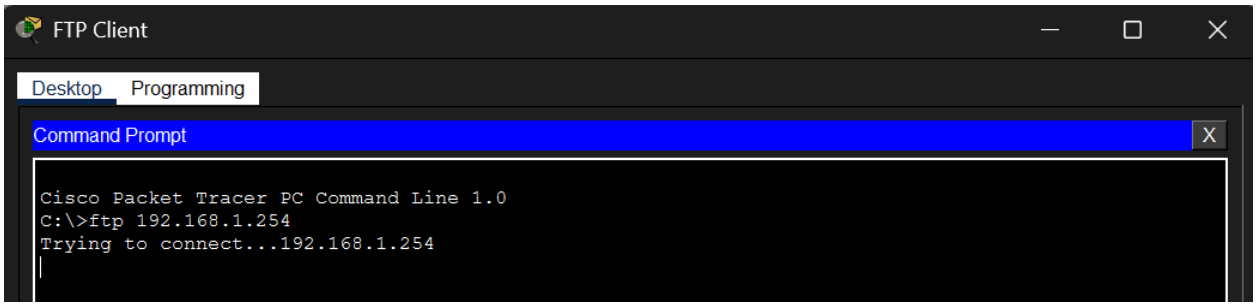


Step 3: Generate FTP traffic.

a. Click FTP Client and open the Command Prompt from the desktop

b. Enter the ftp 192.168.1.254 command. PDUs will appear in the simulation window.

c. Minimize, but do not close, the FTP Client configuration window.

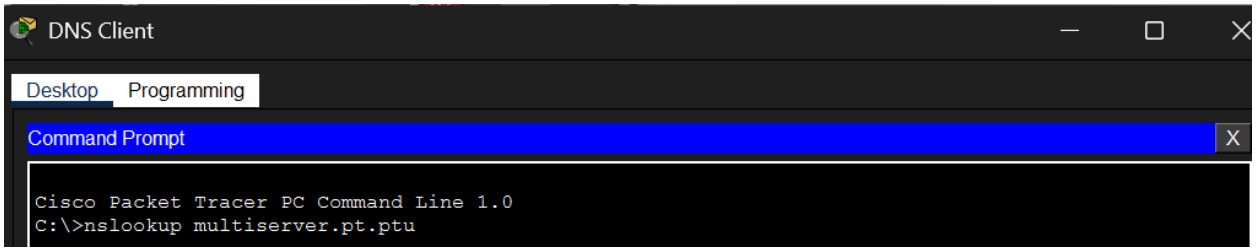


Step 4: Generate DNS traffic.

a. Click DNS Client and open the Command Prompt.

b. Enter the nslookup multiserver.pt.ptu command. A PDU will appear in the simulation window.

c. Minimize, but do not close, the DNS Client configuration window.



Step 5: Generate Email traffic.

a. Click E-Mail Client and open the E Mail tool from the Desktop.

b. Click Compose and enter the following information:

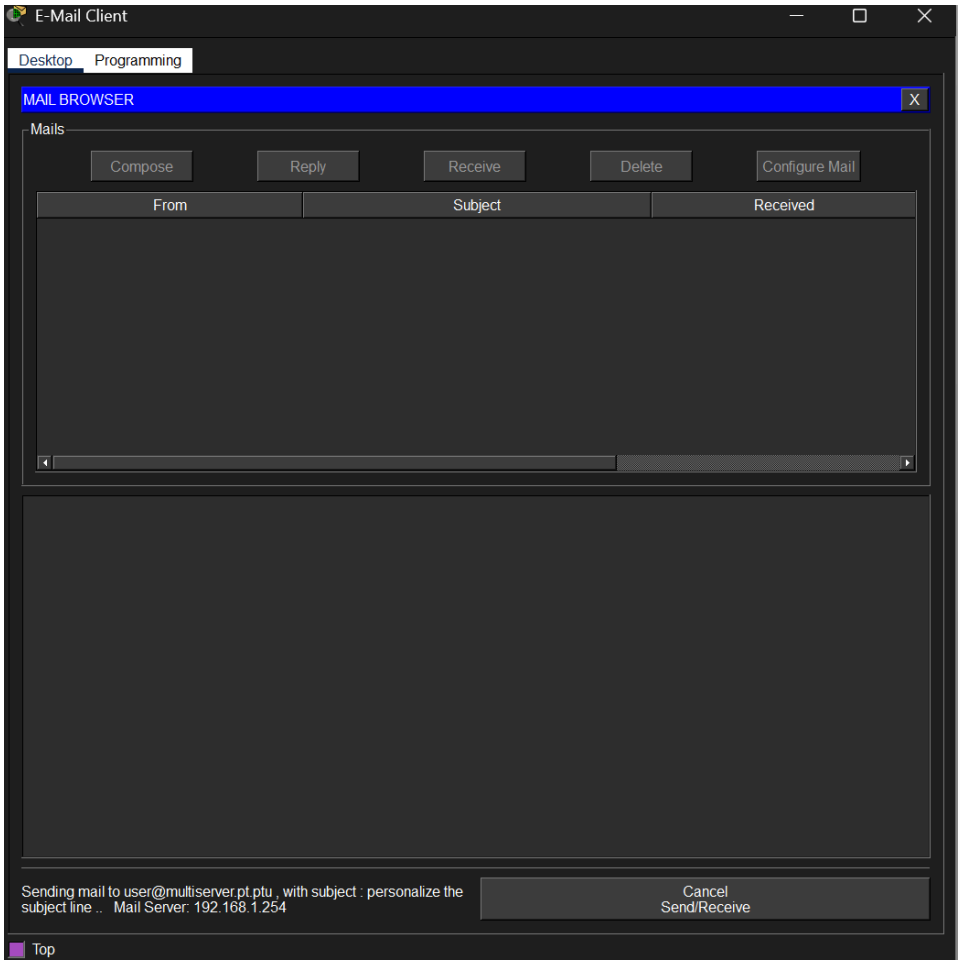
1) To: user@multiserver.pt.ptu

2) Subject: personalize the subject line

3) E-Mail Body: personalize the Email

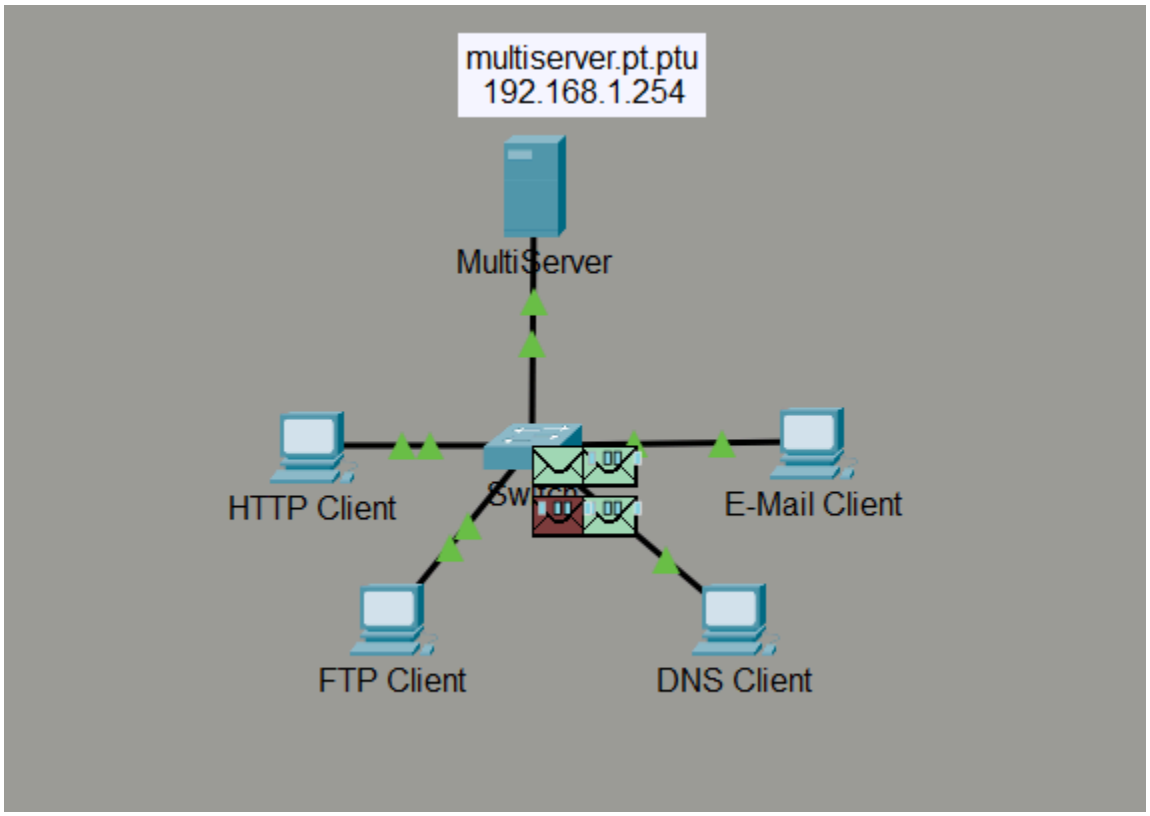
c. Click Send.

d. Minimize, but do not close, the E-Mail Client configuration window.

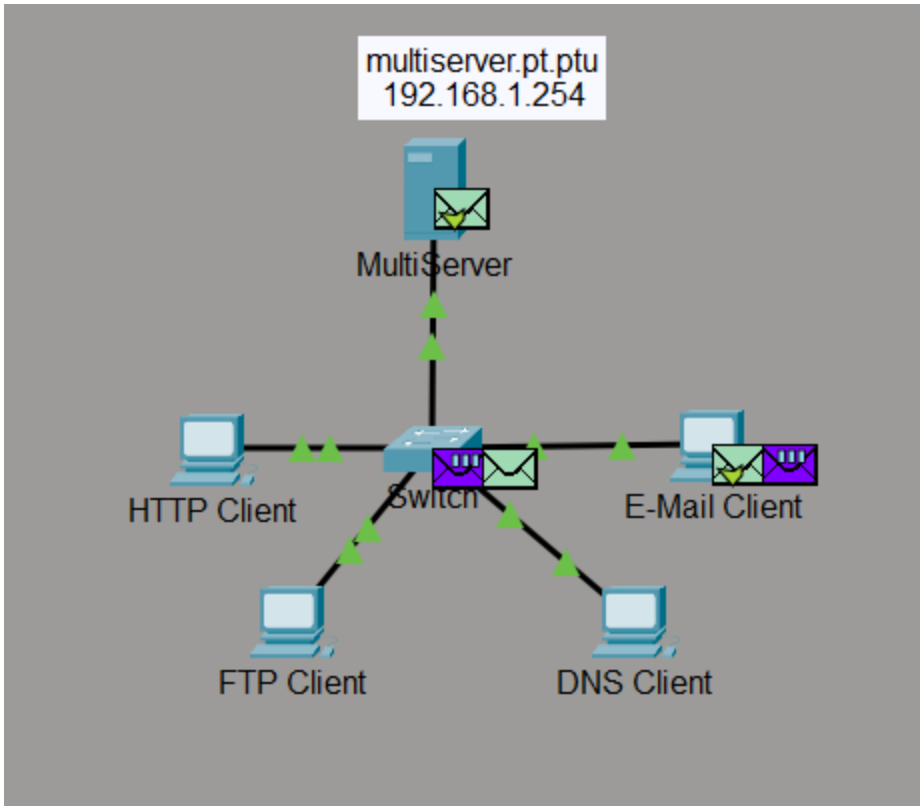


Step 6: Verify that the traffic is generated and ready for simulation.
There should now be PDU entries in the simulation panel for each of the client computers.

Step 7: Examine multiplexing as the traffic crosses the network.
You will now use the Capture/Forward button in the Simulation Panel to observe the different protocols
travelling on the network.
Note: The Capture/Forward button ‘ >| ‘ is a small arrow pointing to the right with a vertical bar next to it.
a. Click Capture/Forward once. All of the PDUs travel to the switch.



b. Click Capture/Forward six times and watch the PDUs from the different hosts as they travel on the



network. Note that only one PDU can cross a wire in each direction at any given time.

Questions:

What is this called?

Type your answers here.

A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different

colors?

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine HTTP traffic as the clients communicate with the server.

a. Click Reset Simulation.

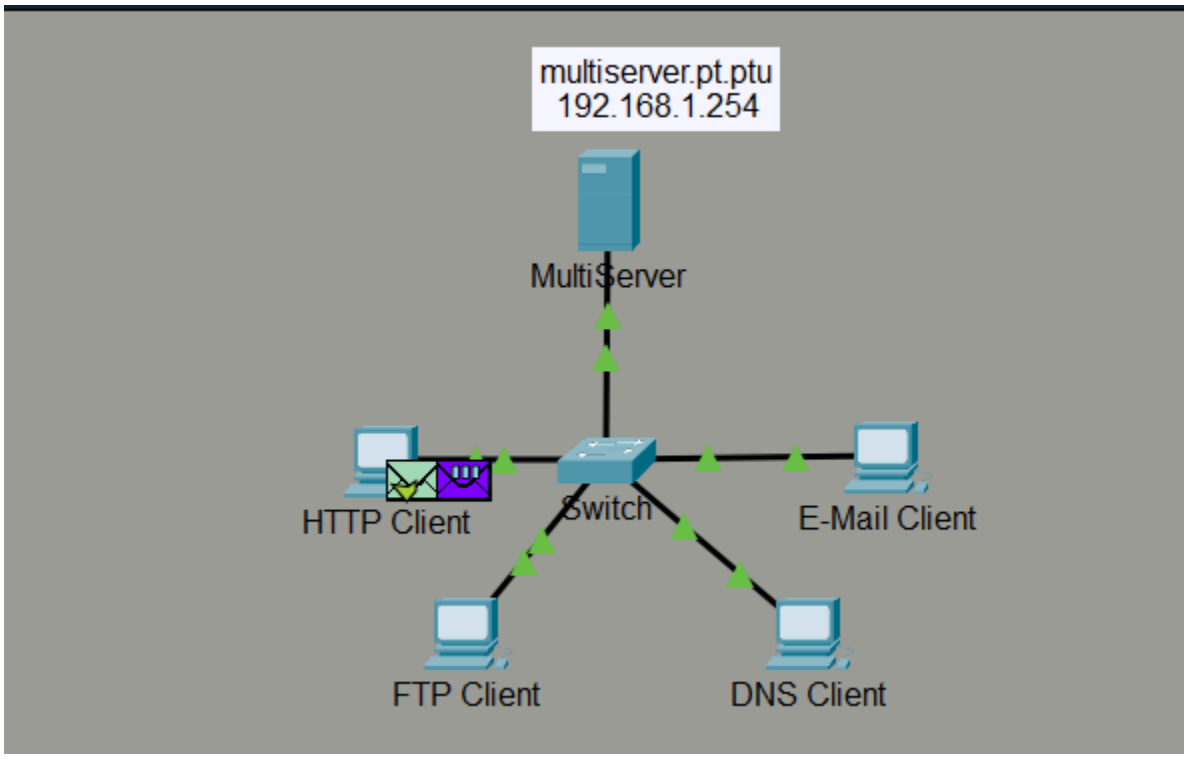
b. Filter the traffic that is currently displayed to only HTTP and TCP PDUs. To filter the traffic that is currently
displayed:

1) Click Edit Filters and toggle the Show All/None button.

2) Select HTTP and TCP. Click the red “x” in the upper right-hand corner of the Edit Filters box to close
it. Visible Events should now display only HTTP and TCP PDUs.

c. Open the browser on HTTP Client and enter 192.168.1.254 in the URL field. Click Go to connect to the
server over HTTP. Minimize the HTTP Client window.

d. Click Capture/Forward until you see a PDU appear for HTTP. Note that the color of the envelope in the



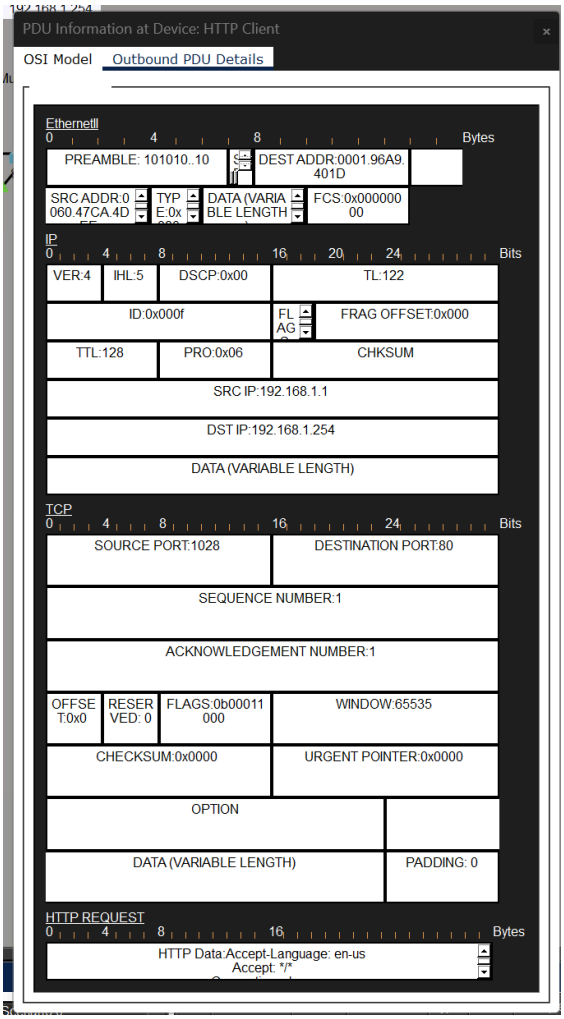
topology window matches the color code for the HTTP PDU in the Simulation Panel.

Question:

Why did it take so long for the HTTP PDU to appear?

-.Actual HTTP data can only be sent after the TCP three-way handshake consisting of SYN, SYN-ACK, and ACK is complete.

e. Click the PDU envelope to show the PDU details. Click the Outbound PDU Details tab and scroll down
to the second to the last section.



What is the section labeled?
-TCP
Are these communications considered to be reliable?
-Yes, because the transport layer protocol being used is TCP which was specifically designed to provide a reliable delivery service.
Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.

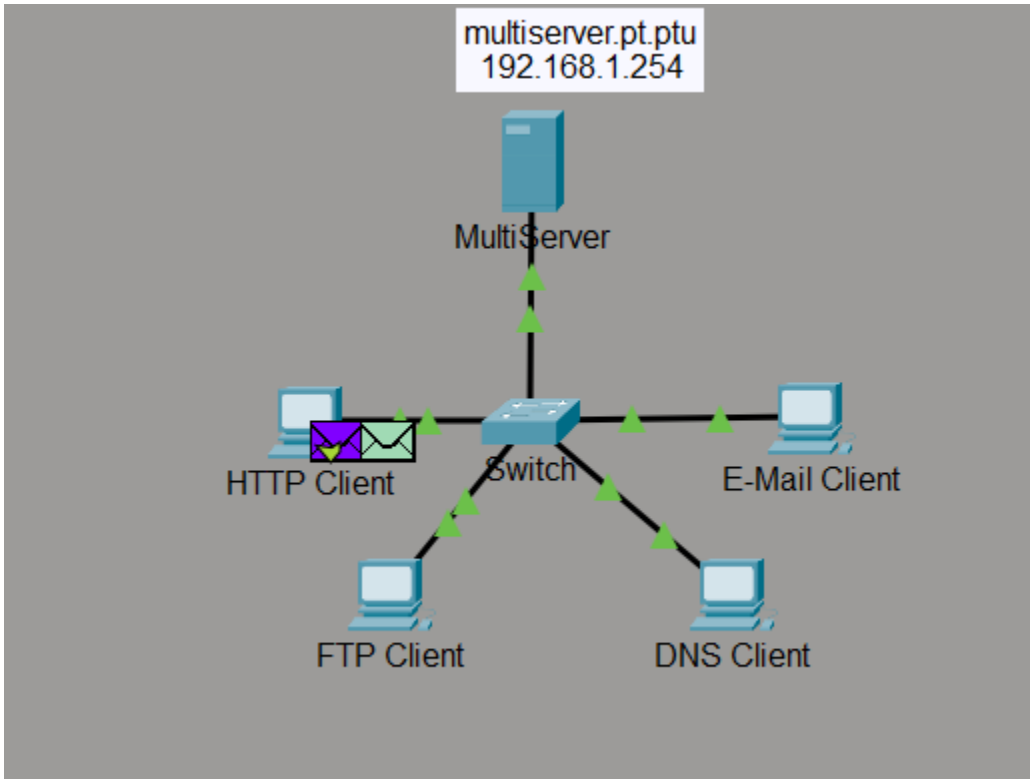
f. Look at the value in the Flags field, which is located next to the Window field. The values to the right of
the “b” represent the TCP flags that are set for this stage of the data conversation. Each of the six places
corresponds to a flag. The presence of a “1” in any place indicates that the flag is set. More than one flag
can be set at a time. The values for the flags are shown below.

Flag Place	6	5	4	3	2	1
Value	URG	ACK	PSH	RST	SYN	FIN

Question:
Which TCP flags are set in this PDU?

- In this PDU, only the **SYN** (Synchronize) flag is set.

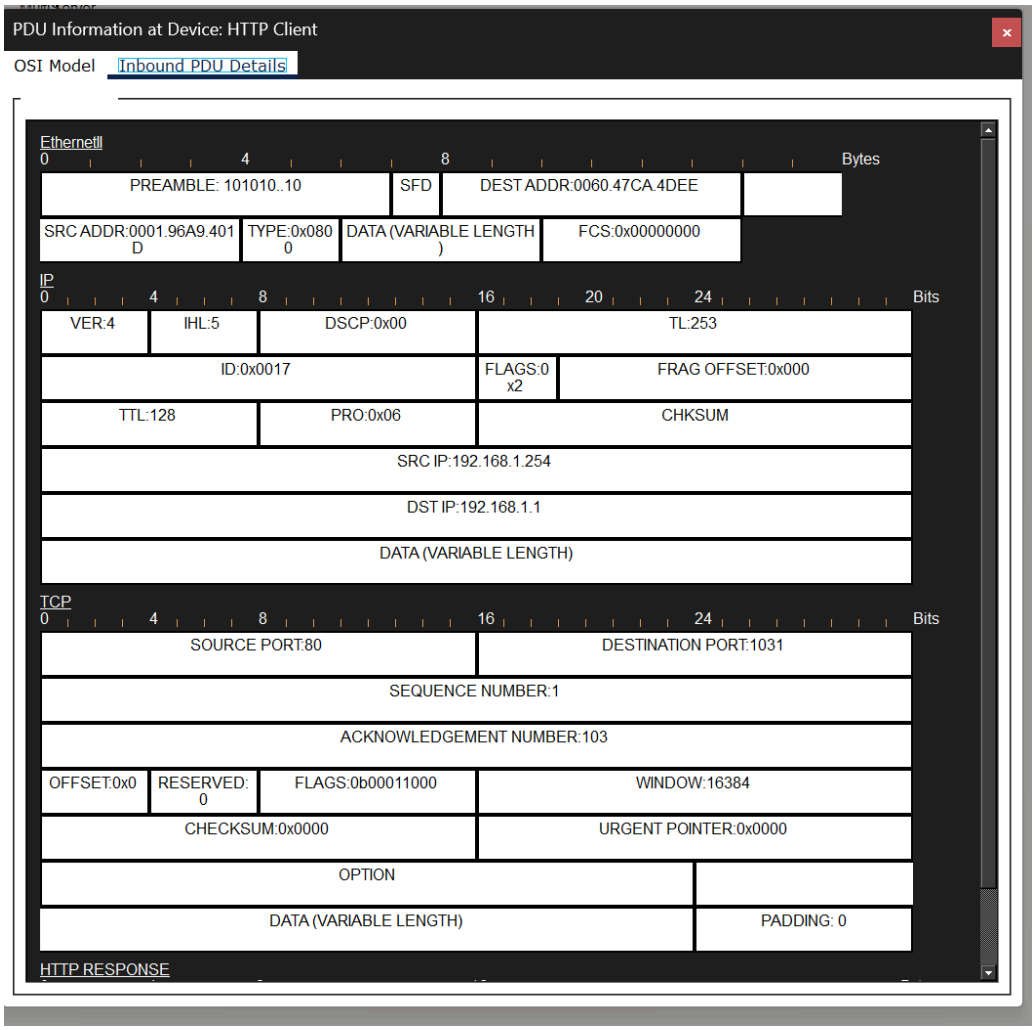
g. Close the PDU and click Capture/Forward until a PDU with a checkmark returns to the HTTP Client.



h. Click the PDU envelope and select Inbound PDU Details.
Question:
How are the port and sequence numbers different than before?

- The response swaps the ports (Source 80, Destination 1025) and increases the Sequence and Acknowledgment numbers to track the communication.

i. Click the HTTP PDU which HTTP Client has prepared to send to MultiServer. This is the beginning of

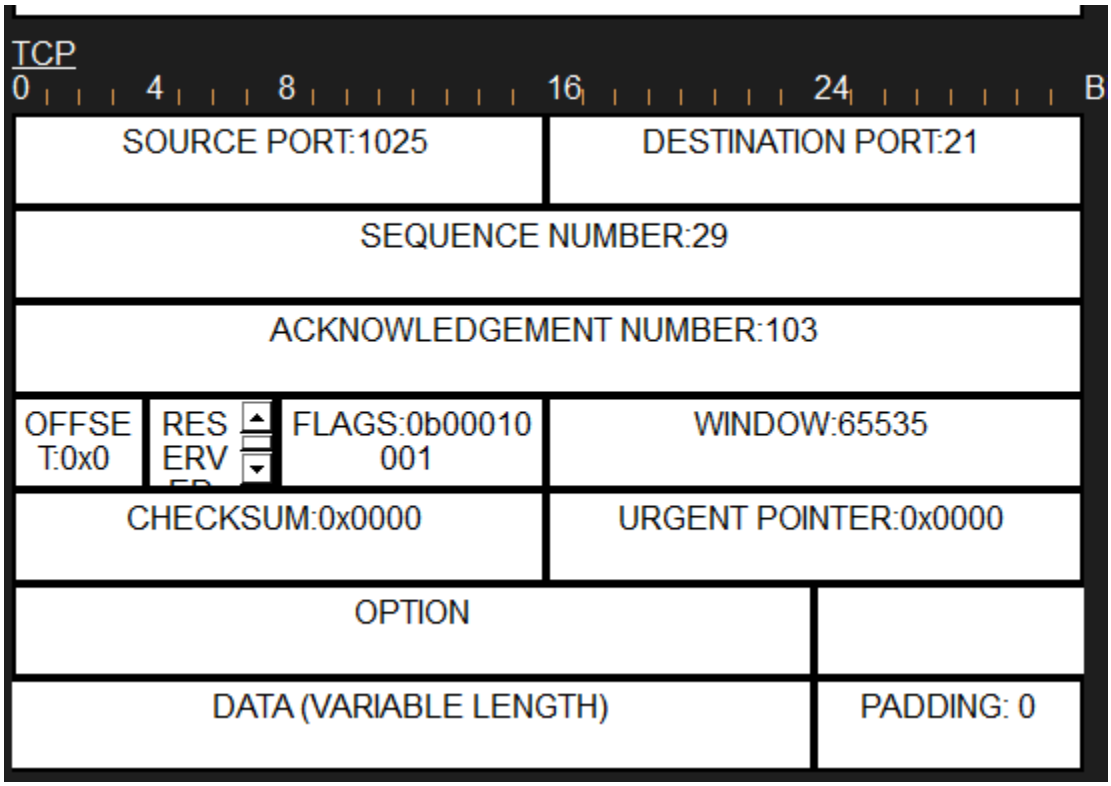


the HTTP communication. Click this second PDU envelope and select Outbound PDU Details.
Question:
What information is now listed in the TCP section? How are the port and sequence numbers different
from the previous two PDUs?

- This PDU carries the actual HTTP GET payload. While the source and destination ports remain constant to maintain the session, the sequence and acknowledgment numbers have been incremented to track the data bytes being exchanged.

j. Reset the simulation.

Step 2: Examine FTP traffic as the clients communicate with the server.
a. Open the command prompt on the FTP Client desktop. Initiate an FTP connection by entering ftp
192.168.1.254.
b. In the Simulation Panel, change Edit Filters to display only FTP and TCP.
c. Click Capture/Forward. Click the second PDU envelope to open it.
Click the Outbound PDU Details tab and scroll down to the TCP section.
Question:
Are these communications considered to be reliable?
Type your answers here.
d. Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.



Question:
What is the value in the flag field?

- SRC PORT: 1025, DEST PORT: 21, SEQUENCE NUM: 1, ACK NUM: 53

e. Close the PDU and click Capture/Forward until a PDU returns to the FTP Client with a checkmark.
f. Click the PDU envelope and select Inbound PDU Details.
Question:
How are the port and sequence numbers different than before?

- Port roles are reversed in the response packet. The Acknowledgment number is calculated by incrementing the sender's Sequence Number by one.

g. Click the Outbound PDU Details tab.

Question:

How are the port and sequence numbers different from the previous results?

- Port assignments remain constant, while the Sequence Number increments to reflect the progression of the data stream.

h. Close the PDU and click Capture/Forward until a second PDU returns to the FTP Client. The PDU is a
different color.

i. Open the PDU and select Inbound PDU Details. Scroll down past the TCP section.

Question:

What is the message from the server?

- `220- Welcome to PT Ftp server`

j. Click Reset Simulation.

Step 3: Examine DNS traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to create DNS traffic.

b. In the Simulation Panel, change Edit Filters to display only DNS and UDP.

c. Click the PDU envelope to open it.

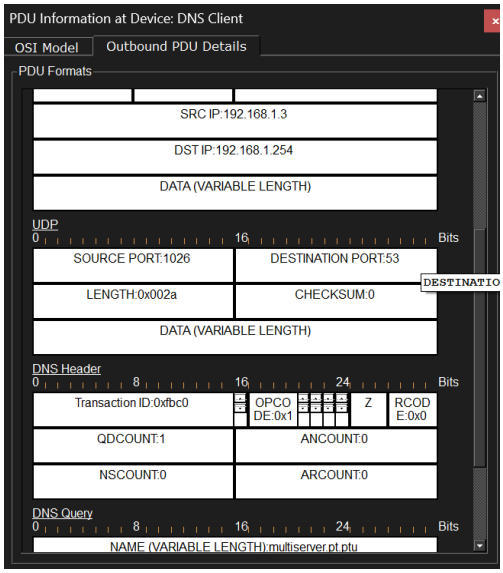
What is the Layer 4 protocol?

- The Layer 4 protocol is **UDP**

Are these communications considered to be reliable?

- This communication is **not** considered to be reliable. In networking, **UDP** is a connectionless and unreliable transport protocol because the header contains only Source Port, Destination Port, Length, and Checksum. It does not have a field to acknowledge (ACK) that the data was received.

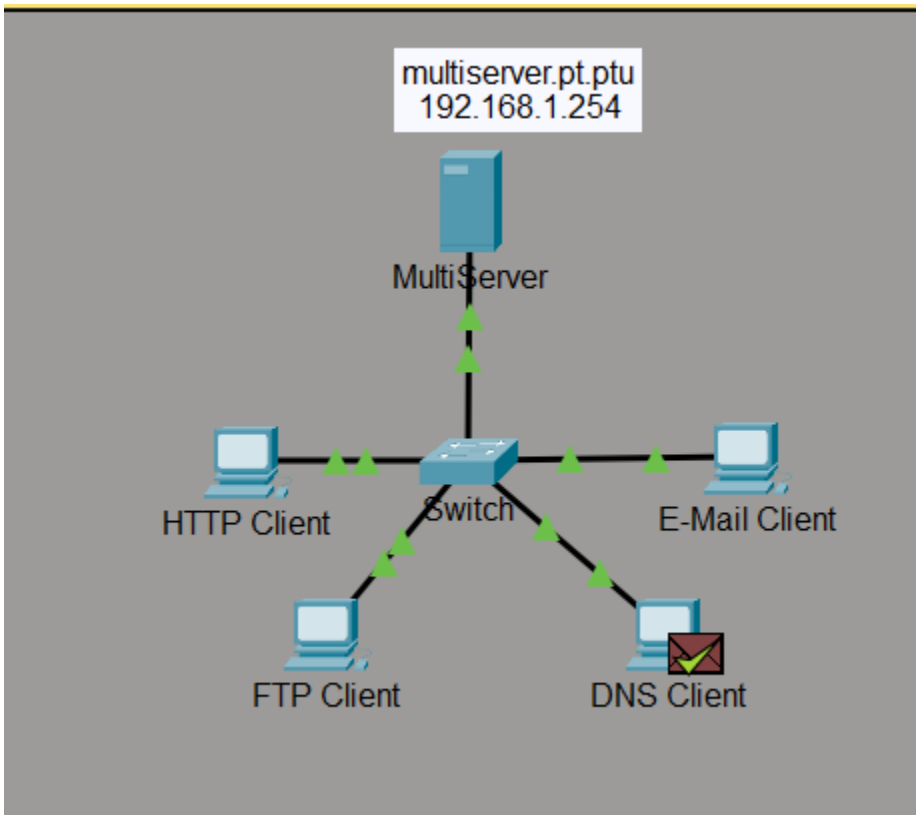
e. Open the Outbound PDU Details tab and find the UDP section of the PDU formats. Record the SRC PORT and DEST PORT values.



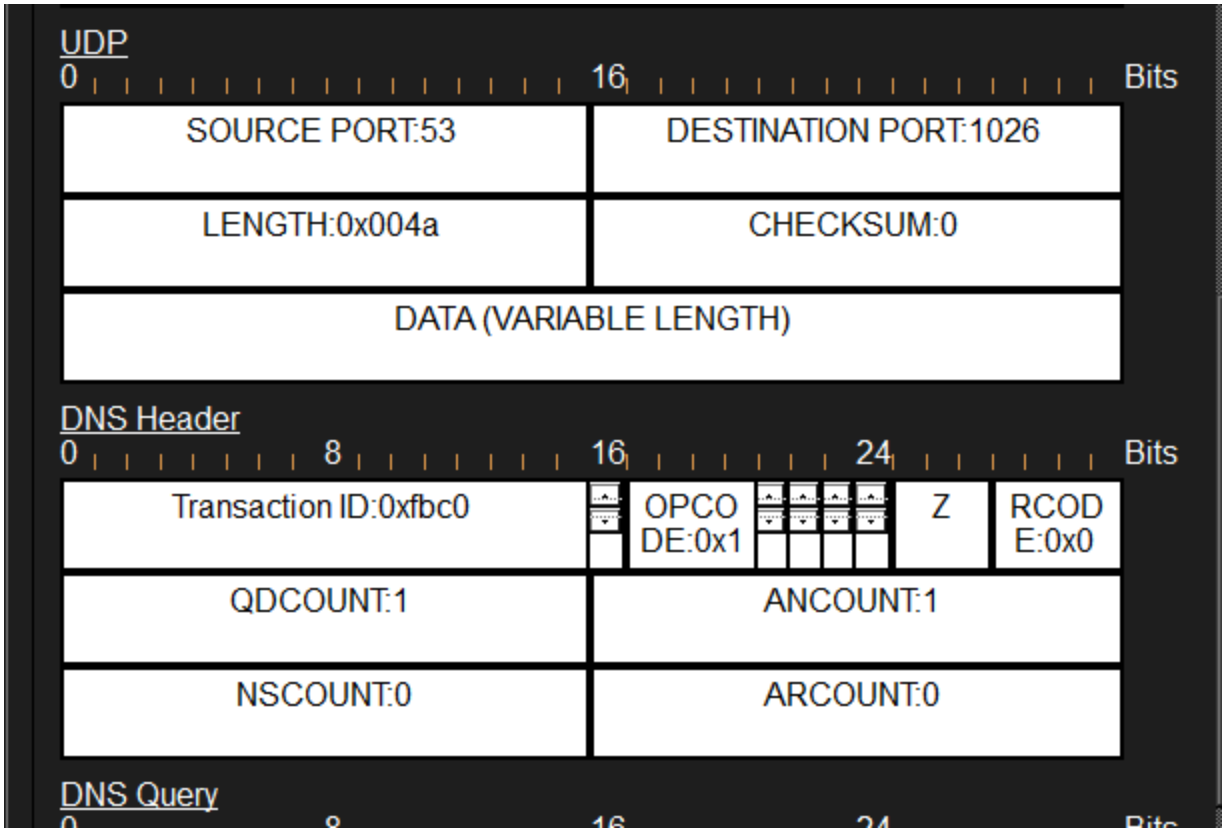
Why are there no sequence and acknowledgement numbers?

- Because **the communication is using the UDP** at Layer 4. Unlike TCP, which uses those fields to ensure reliability and order, UDP is a connectionless protocol designed for speed and efficiency.

f. Close the PDU and click Capture/Forward until a PDU with a check mark returns to the DNS Client.



g. Click the PDU envelope and select Inbound PDU Details.



Question:
How are the port and sequence numbers different than before?

- The source and destination ports are swapped (Source: 53, Destination: 1025), and there are no sequence numbers because DNS uses the connectionless UDP protocol.

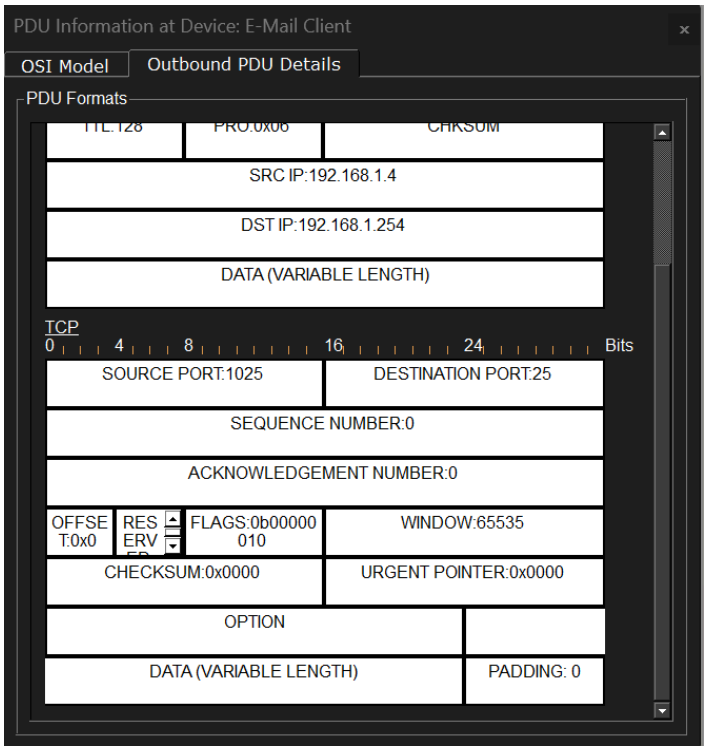
What is the last section of the PDU called? What is the IP address for the name multiserver.pt.ptu?

- The last section of the PDU is called **DNS Answer**. According to this section, the IP address for the name multiserver.pt.ptu is **192.168.1.254**

h. Click Reset Simulation.

Step 4: Examine email traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to send an email to user@multiserver.pt.ptu.
b. In the Simulation Panel, change Edit Filters to display only POP3, SMTP and TCP.
c. Click the first PDU envelope to open it.
d. Click the Outbound PDU Details tab and scroll down to the last section



Questions:
What transport layer protocol does email traffic use?

- TCP

Are these communications considered to be reliable?

- Yes because TCP provides reliable, ordered delivery by using sequence numbers to reassemble data correctly and acknowledgments to ensure no packets are lost.

e. Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values. What is the flag field
Value?

- SRC PORT: 1025, DEST PORT: 80, SEQUENCE NUM: 0, ACK NUM: 0

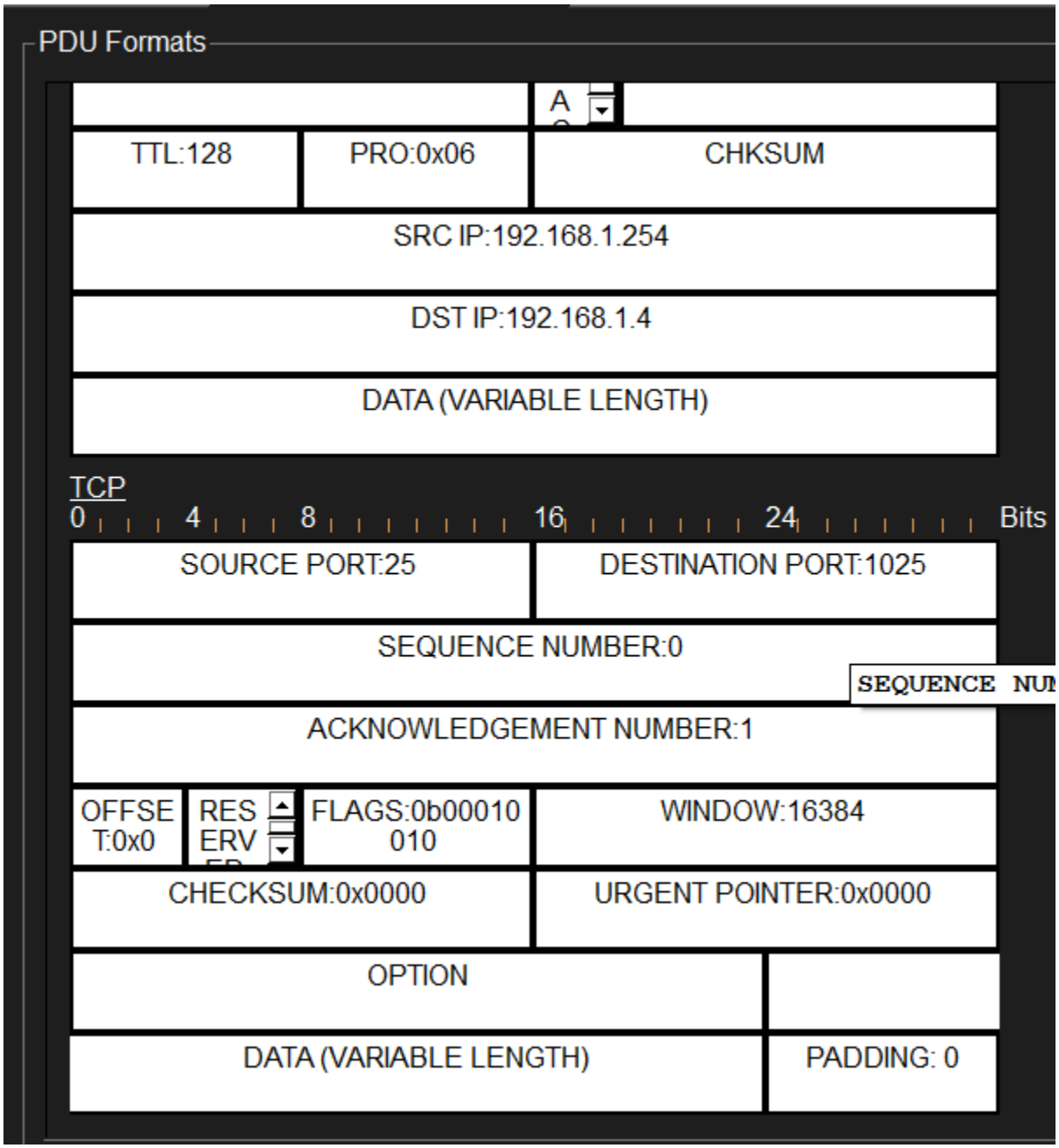
f. Close the PDU and click Capture/Forward until a PDU returns to the E-Mail Client with a checkmark.
g. Click the TCP PDU envelope and select Inbound PDU Details.

Question:
How are the port and sequence numbers different than before?

- The source and destination ports have swapped roles; the Source Port is now 25 and the Destination Port is 1025.

- The Sequence Number remains 0, but the Acknowledgment Number has increased to 1 to confirm receipt of the previous SYN packet.

h. Click the Outbound PDU Details tab.



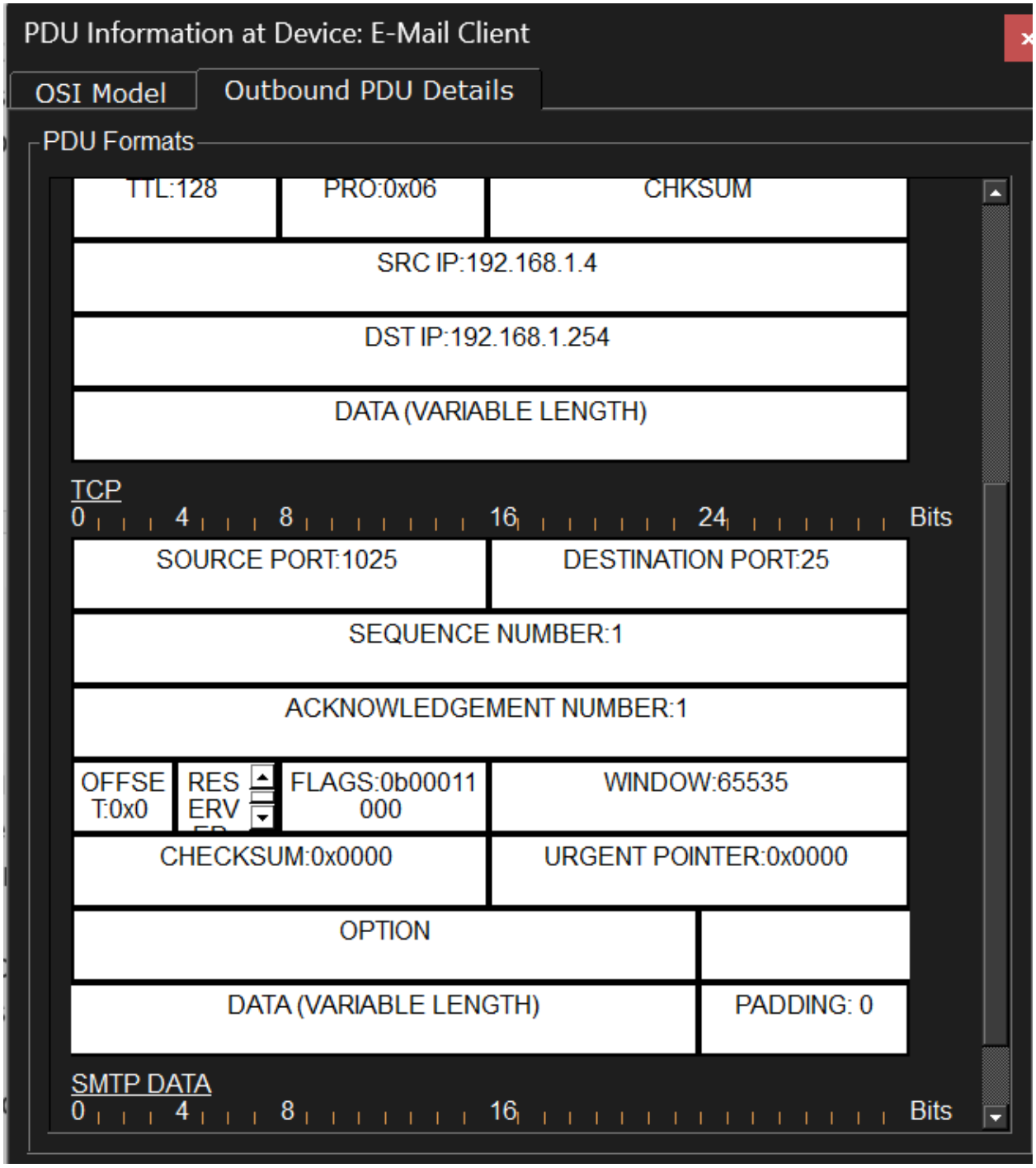
Question:

How are the port and sequence numbers different from the previous two results?

- The roles have swapped again compared to the previous inbound PDU, the Source Port is back to 1025 and the Destination Port is 25.

- Both values are now set to 1. The Sequence Number increased to 1 to match the acknowledgment received from the server, and the Acknowledgement Number remains 1 to confirm the server's previous SYN/ACK segment.

i. There is a second PDU of a different color that E-Mail Client has prepared to send to MultiServer. This



is the beginning of the email communication. Click this second PDU envelope and select Outbound PDU

Details.
Questions:
How are the port and sequence numbers different from the previous two PDUs?

- The ports remain the same as the final handshake step, with a Source Port of 1025 and a Destination Port of 25.
- Both the Sequence Number and Acknowledgment Number are set to 1.

What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

- **TCP Port 25: SMTP**, which is used for sending or relaying email between clients and servers.
- **TCP Port 110: POP3**, which is used by email clients to retrieve or download emails from a server.

Poonyawee tankam 673380282-8 sec1